

BIOL 40C

Class 10

Aldosterone & Acid-Base Balance

Oscar's Case | Midterm Setup

Wednesday, May 6, 2026 | 10:00–11:50 AM | Room 8403

Giorgio Lagna, Ph.D. — Foothill College

Today's Roadmap

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CONCEPT

Block 1

Welcome + Oscar — Step 1

10:00–10:25

Block 2

Aldosterone, Sodium & Potassium + Step 2 (Diagnosis)

10:25–10:55

Block 3

Acid-Base Balance + ABG Reading + Step 3 (Cellular)

11:05–11:35

Block 4

Step 4 (Treatment) + Integration + Midterm

11:35–11:50

Bridge from Monday

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CONCEPT

Monday: how does the body decide how much WATER and how much SODIUM to keep?

Today: how does the body decide how ACIDIC the blood should be?

Same answer in shape: instant buffers, fast lungs, slow kidneys.

The hierarchy of speeds is the same as Monday's. The rest is chemistry.

And today: ONE patient, ONE hormone, FOUR lab findings — the most elegant case of the week.

KC Quiz 4 — Top Miss

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A&P REVIEW

Predicted top miss:

"What is the primary disturbance in an ABG showing pH 7.30, PCO_2 25, HCO_3^- 12?"

This is a layered question — you need both an acidosis read AND a primary-driver read.

We'll teach the 30-second algorithm in Block 3 today.

Meet Oscar

Oscar, age 40, comes in for his annual checkup.

"Doc, I've been doing everything you said — exercise, diet, low salt — but my pressure isn't coming down. I've tried three blood-pressure meds and I'm still high. Also, I sometimes feel a strange muscle weakness, kind of like cramping."

On exam: BP elevated. He has been treatment-resistant for several visits.

Think-Pair-Share

ACTIVITY

With your team, take 2 minutes:

- What conditions might Oscar be suffering from?
- What additional information would help you narrow the diagnosis?

Things to weigh:

- *"Treatment-resistant hypertension" is a clinical phrase that points somewhere specific.*
- *Muscle weakness + hypertension: what electrolyte connects those two?*

Open Your Worksheet — AA-3: Oscar's Case

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ACTIVITY

We'll work Oscar's case live in class, across 4 steps, in your team.

Open AA-3 in Canvas → make a copy of the worksheet → log your team's reasoning as we go.

- 1** Open the AA-3 worksheet on Canvas now (laptop or printout)
- 2** Make your own copy (Google Doc or local download)
- 3** Capture your team's reasoning at each of the 4 steps as we work through them today
- 4** Finish the write-up after class. Submit as PDF on Canvas by Sat May 9, 11:59 PM

If you didn't finish AA-2 (Reyna's Case) yet, today's framework will help; both cases are due Saturday.

Block 2

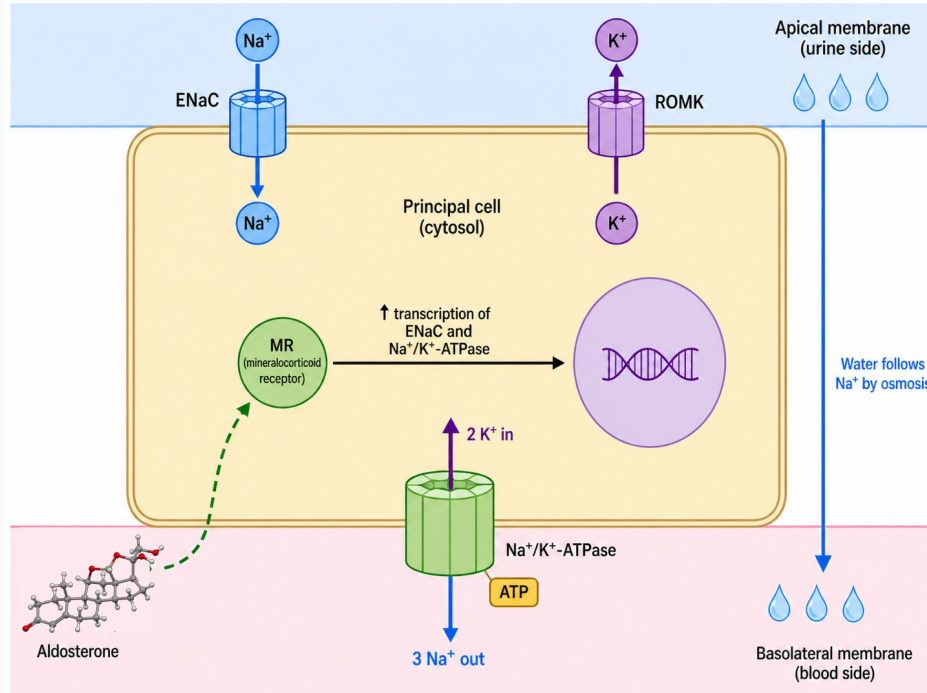
Aldosterone — The Sodium-Potassium Switch

Building the framework before we look at Oscar's labs

Sodium Balance — RAAS in One Picture

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CONCEPT



Amerman Fig. 25.8 — RAAS feedback

Aldosterone reabsorbs Na⁺.

In the principal cells of the collecting duct:

- Na⁺ IN (apical ENaC)
- Na⁺ pumped to blood (Na/K ATPase)
- Water follows Na⁺
- K⁺ secreted into urine
- (H⁺ secreted into urine by the intercalated cell)

Na⁺ IN ↔ K⁺ + H⁺ OUT

The Aldosterone Exchange

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CONCEPT

Na^+ IN $\leftrightarrow$ K^+ + H^+ OUT

Sodium reabsorbed; potassium and hydrogen secreted into urine.

More Na^+ reabsorbed $\rightarrow$

- Water follows $\rightarrow$ ECF expands
- Blood pressure $\uparrow$
- Plasma $\text{Na}^+ \approx$ unchanged
(water dilutes it back)

More K^+/H^+ secreted $\rightarrow$

- Plasma $\text{K}^+ \downarrow$
- Plasma $\text{H}^+ \downarrow \rightarrow$ blood pH $\uparrow$
- If sustained: METABOLIC ALKALOSIS

Predict Each Arrow

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CHECK

If aldosterone goes UP, what happens to:

Plasma Na^+ ?

Plasma K^+ ?

Plasma H^+ (and pH)?

Blood pressure?

Hold these predictions; we'll match them to a real patient in a moment.

Vote on Poll Everywhere

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CHECK



Vote at:

PollEv.com/glagna

Or scan the QR code →

Live results display
once everyone votes.

Q: If aldosterone goes UP, what happens to plasma Na⁺, K⁺, H⁺/pH, and blood pressure? (Polls Q1a-Q1d)

BIOL40C-W4-C10-Q1a: If aldosterone goes UP, what happens to PLASMA Na^+ ?



Substantially increases

0%

Substantially decreases

0%

Roughly unchanged (water follows)

0%

BIOL40C-W4-C10-Q1b: If aldosterone goes UP, what happens to PLASMA K+?



Increases

0%

Decreases

0%

Roughly unchanged

0%

BIOL40C-W4-C10-Q1c: If aldosterone goes UP, what happens to BLOOD pH?



Rises (alkalosis)

0%

Falls (acidosis)

0%

Roughly unchanged

0%

BIOL40C-W4-C10-Q1d: If aldosterone goes UP, what happens to BLOOD PRESSURE?



Rises

0%

Falls

0%

Roughly unchanged

0%

Aldosterone Up: All Four Answers

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Plasma analyte / metric	Direction
Plasma Na ⁺	Roughly unchanged (water follows)
Plasma K ⁺	Decreases
Blood H ⁺ / pH	H ⁺ decreases → pH rises (alkalosis)
Blood pressure	Rises

All four are the same hormone — Na⁺ in, K⁺ + H⁺ out, water (and BP) follow.

Oscar's Test Results

STEP

You sent Oscar for blood tests. Below are his most relevant results:

Blood Test	Oscar's Result
Plasma Potassium (K^+)	LOW
Aldosterone-to-Renin Ratio (ARR)	HIGH
Blood pH	HIGH

Plus: BP elevated. Treatment-resistant. Muscle weakness reported.

Interpret These Results

ACTIVITY

With your team (5 minutes):

1

What is being measured in each test, and what is each value telling you?

2

What prior knowledge do you have about renin, aldosterone, K^+ , and pH that links them in a single mechanism?

3

What ONE hormone could explain all four findings (low K^+ , high ARR, high pH, high BP)?

4

Why is the renin LOW even though the BP is HIGH? (The negative-feedback loop is broken — how?)

Reason it out from what you've learned. Don't ask the instructor for the answer yet.

Break

10 minutes — back at 11:00

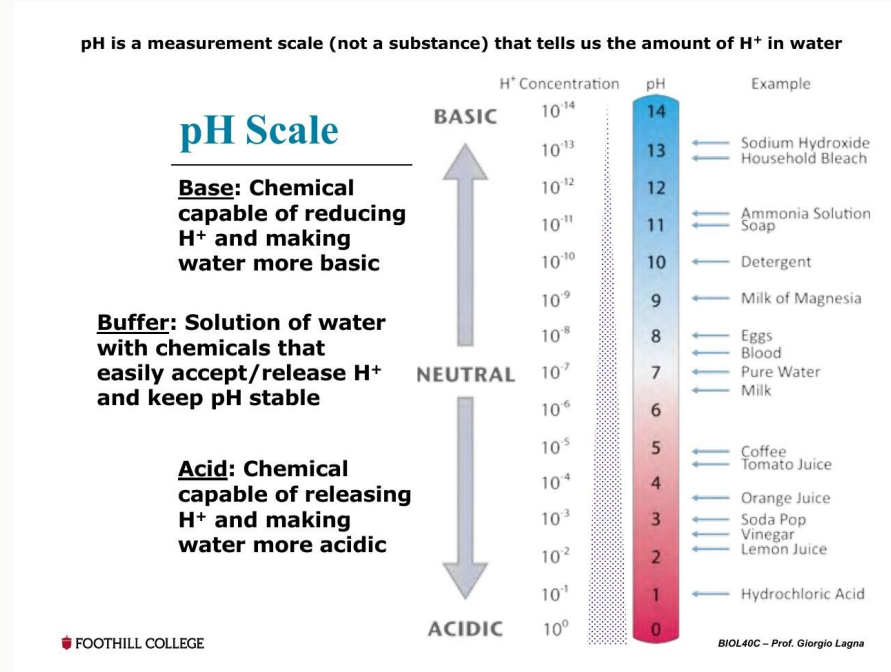
Block 3

Acid-Base Balance

...and reading Oscar's ABG in 30 seconds

pH — A Measurement Scale, Not a Substance

CONCEPT



Normal arterial pH: 7.35–7.45. Below 6.8 or above 7.8 is incompatible with life.

Three Lines of pH Defense

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CONCEPT

Instant

Chemical buffers

bicarbonate, phosphate, proteins (Hgb, albumin)

Exhaustible

Minutes

Respiratory compensation

change ventilation → change PCO_2

Partial

Days

Renal compensation

excrete fixed acids; reclaim/generate HCO_3^-

Complete

Same architecture as Monday's water/sodium hierarchy.

The Bicarbonate Equation

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CONCEPT



LUNGS control the LEFT side

Hyperventilate → blow off CO_2 → ↓ left side → drives reaction LEFT → ↓ H^+ → pH ↑

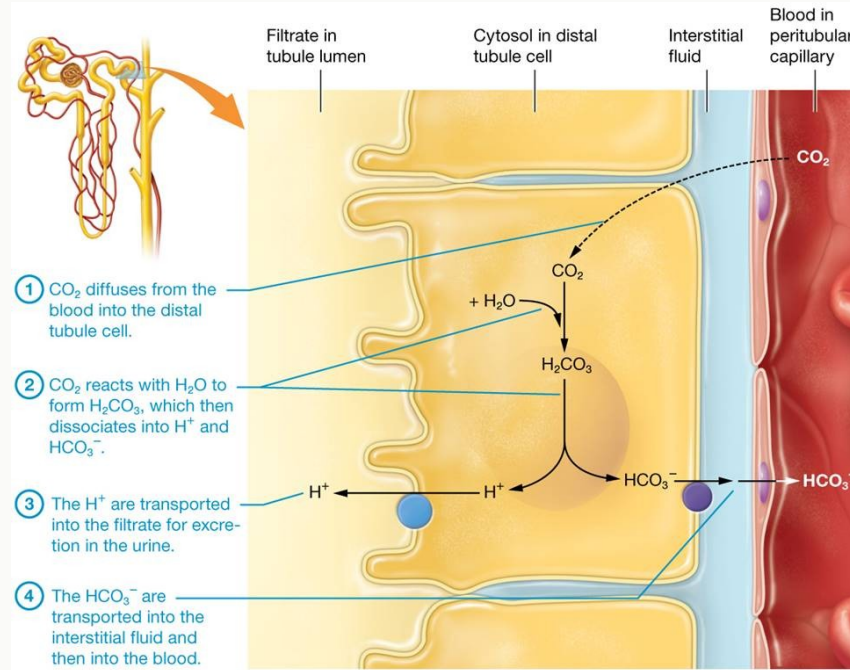
KIDNEYS control the RIGHT side

Excrete H^+ , generate new HCO_3^- → ↑ right side → drives reaction RIGHT → buffers acid load

How the Kidney Excretes H^+ and Generates HCO_3^-

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CONCEPT



Amerman Fig. 25.10 — distal tubule cell exchanges H^+ for HCO_3^-

The Four Primary Disturbances

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CONCEPT

	Low pH (ACIDOSIS)	High pH (ALKALOSIS)
PCO₂ driver (RESPIRATORY)	Respiratory acidosis (↑ PCO ₂) e.g., COPD, hypoventilation	Respiratory alkalosis (↓ PCO ₂) e.g., panic, hyperventilation
HCO₃⁻ driver (METABOLIC)	Metabolic acidosis (↓ HCO ₃ ⁻) e.g., DKA, uremia, diarrhea	Metabolic alkalosis (↑ HCO ₃ ⁻) e.g., vomiting, ↑ aldosterone

TOOL



Open on your laptop:

Acid-Base 3D Explorer

- Rotate the $\text{pH} \times \text{PCO}_2 \times \text{HCO}_3^-$ space
- Click any region for the disorder
- Plot any custom ABG live
- Use it for midterm review

Link in Assignments and Activities for Week 4, Class Materials: Session 10

https://glagna.github.io/AP40/AcidBase_3D_Explorer.html

The 4 disorders aren't 4 boxes; they're 4 corners of a 3D space. Mixed disorders live in between.

The 30-Second ABG Algorithm

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CONCEPT

1

Look at pH

$<7.35 = \text{acidosis}$ / $>7.45 = \text{alkalosis}$

2

Look at PCO_2

normal 35–45

3

Look at HCO_3^-

normal 22–26

4

Match them up

pH & PCO_2 opposite $\rightarrow$ respiratory / pH & HCO_3^- same $\rightarrow$ metabolic

Drill this; it's on the midterm.

Practice — Classify These Four ABGs

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ACTIVITY

ABG 1: pH 7.28 / PCO_2 60 / HCO_3^- 28

ABG 2: pH 7.55 / PCO_2 25 / HCO_3^- 22

ABG 3: pH 7.18 / PCO_2 22 / HCO_3^- 8

ABG 4: pH 7.48 / PCO_2 45 / HCO_3^- 32

90 seconds with your neighbor. Use the algorithm.

Vote on Poll Everywhere

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Vote at:

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Or scan the QR code →

Live results display
once everyone votes.

Q: Classify each of the four ABGs above. (Polls Q2a-Q2d, one per ABG)

BIOL40C-W4-C10-Q2a: Classify ABG 1: pH 7.28 / PCO₂ 60 / HCO₃⁻ 28



Respiratory acidosis

0%

Respiratory alkalosis

0%

Metabolic acidosis

0%

Metabolic alkalosis

0%

BIOL40C-W4-C10-Q2b: Classify ABG 2: pH 7.55 / PCO₂ 25 / HCO₃⁻ 22



Respiratory acidosis

0%

Respiratory alkalosis

0%

Metabolic acidosis

0%

Metabolic alkalosis

0%

BIOL40C-W4-C10-Q2c: Classify ABG 3: pH 7.18 / PCO₂ 22 / HCO₃⁻ 8



Respiratory acidosis

0%

Respiratory alkalosis

0%

Metabolic acidosis

0%

Metabolic alkalosis

0%

BIOL40C-W4-C10-Q2d: Classify ABG 4: pH 7.48 / PCO₂ 45 / HCO₃⁻ 32



Respiratory acidosis

0%

Respiratory alkalosis

0%

Metabolic acidosis

0%

Metabolic alkalosis

0%

ABG 1: pH 7.28 / PCO_2 60 / HCO_3^- 28

Respiratory acidosis (chronic, with renal compensation)

ABG 2: pH 7.55 / PCO_2 25 / HCO_3^- 22

Respiratory alkalosis (acute, hyperventilation)

ABG 3: pH 7.18 / PCO_2 22 / HCO_3^- 8

Metabolic acidosis with respiratory compensation (this is what Santiago looked like)

ABG 4: pH 7.48 / PCO_2 45 / HCO_3^- 32

Metabolic alkalosis — and this matches Oscar.

Summary of Acid-Base Disorders

A&P REVIEW

Table 25.1 Summary of Acid-Base Disorders

Type of Disorder	Characteristic Blood Gas Findings (without Compensation)	Causes	Main Methods of Compensation
Respiratory acidosis	- Decreased pH - Increased P_{CO_2}	- Blockage of air passages in the lungs - Decreased gas exchange in the alveoli - Decreased ventilation from brainstem dysfunction	- Hyperventilation to remove CO_2 - Retention of HCO_3^- by the kidneys - Excretion of H^+ by the kidneys
Respiratory alkalosis	- Increased pH - Decreased P_{CO_2}	- Psychological states that cause hyperventilation - Brain tumor or other damage to respiratory centers - Hyperventilation due to high altitudes and low atmospheric P_{O_2}	- Slowed respiratory rate - Excretion of HCO_3^- and retention of H^+ by the kidneys (if alkalosis is physiological rather than psychological)
Metabolic acidosis	- Decreased pH - Decreased HCO_3^-	- Prolonged diarrhea - Excessive production of metabolic acids - Renal failure - Ingestion of acidic drugs or toxins	- Hyperventilation - Retention of HCO_3^- from the kidneys - Excretion of H^+ by the kidneys * (The second and third responses are only possible if renal failure is not the cause of metabolic acidosis.)
Metabolic alkalosis	- Increased pH - Increased HCO_3^-	- Prolonged vomiting - Ingestion of certain diuretics or large doses of oral antacids - Excess aldosterone	- Slowed respiratory rate - Retention of H^+ by the kidneys - Excretion of HCO_3^- by the kidneys

Amerman Table 25.1 — typical findings + compensations + causes

HEALTH EQUITY

Sickle cell disease

- ~100,000 Americans live with SCD
- 1 in 365 Black births in the US; also affects Latinx and South Asian populations
- Originated as a protective mutation against malaria in West Africa

What you'll see in their ABGs

- Chronic compensated metabolic acidosis from hemolysis
- Episodic respiratory acidosis during acute chest crises
- Anion-gap shifts during opioid-related hypoventilation in pain-crisis admits

The framework you just built reads every one of these. Sickle cell research and pain management remain under-resourced — your generation closes that gap.

What's Happening at the Cellular Level?

STEP

Use the framework you just built. With your team:

1

Why would renin be LOW in this condition? (Trace the negative-feedback loop and explain why it's broken.)

2

Why would K^+ be LOW and pH be HIGH? (Hint: think about what aldosterone exchanges in the collecting duct.)

3

How does this condition relate to the high BP and the muscle weakness?

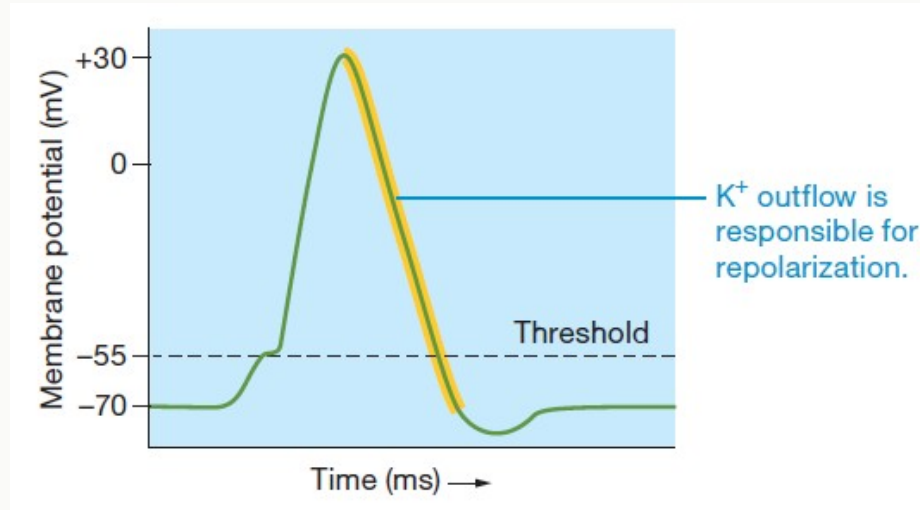
4

Connect the muscle weakness specifically: what does low extracellular K^+ do to a cell's resting membrane potential?

K⁺ Sets the Resting Membrane Potential

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A&P REVIEW



Amerman Fig. 25.6

Low extracellular K⁺ →

- The equilibrium potential for K⁺ becomes MORE negative
- Resting membrane potential hyperpolarizes
- Bigger stimulus needed to fire action potentials
- Muscle: sluggish → weakness
- Cardiac: T-wave flattening, U waves, arrhythmia risk

This is why severe hypokalemia from an aldosterone problem can be fatal.

Quick Poll — Hyperkalemia

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Hyperkalemia (HIGH K^+) does the OPPOSITE of hypokalemia: it depolarizes the cell.

Predict the muscle effect:

A. Weakness

B. Hyperexcitability

C. Both, depending on severity

Vote on Poll Everywhere

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CHECK



Vote at:

PollEv.com/glagna

Or scan the QR code →

Live results display
once everyone votes.

Q: Hyperkalemia (HIGH K⁺) causes which muscle effect? (Poll Q3)

BIOL40C-W4-C10-Q3: Hyperkalemia (HIGH K⁺) causes which muscle effect?



Weakness (cells become less excitable)

0%

Hyperexcitability (cells fire too easily)

0%

Both, depending on severity

0%

Answer: A. Weakness

Hyperkalemia chronically depolarizes excitable cells → voltage-gated Na^+ channels get stuck in their inactivated state → cells can't fire APs → muscle weakness.

Same destination, opposite mechanism:

- HYPOkalemia → hyperpolarizes → harder to reach threshold → weakness
- HYPERkalemia → chronically depolarizes → Na^+ channels inactivated → weakness

BOTH K^+ extremes cause weakness. This is on the midterm.

Block 4

Step 4 — Treatment + Integration + Midterm

Propose a Treatment Plan

ACTIVITY

With your team:

- 1 What are the two main treatment options for Oscar's condition, and how does the choice depend on imaging?
- 2 How would the medical option correct each of his lab abnormalities (K^+ , pH, BP)? Be specific about mechanism.
- 3 Predict Oscar's K^+ , pH, and BP at 4 weeks of treatment.

Hint: there's a class of drugs that BLOCKS the receptor for the hormone in question.

Why Oscar's Diagnosis Took So Long

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HEALTH EQUITY

Common, but missed

- Primary aldosteronism is the most common SECONDARY cause of HTN
- ~5–10% of all HTN
- ~20% of treatment-resistant HTN
- Often missed for 5–10 years before the workup happens

Two equity gaps

- Diagnostic bias: Black and Latinx patients with resistant HTN are more often labeled "non-adherent" than worked up for endocrine causes
- Workup access: ARR screening costs ~\$30 but isn't routinely ordered in under-resourced primary care

Knowing what to look for is the first step in closing the gap. That's what tonight's review buys you.

Weeks 3 + 4 in One Slide

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CONCEPT

Disturbance	Instant	Fast (min-hr)	Slow (hr-day)
Free water deficit	thirst sensors	ADH → free H ₂ O	(kidney = fast here)
Volume deficit	sympathetic	RAAS → Na ⁺ retention	aldosterone
Acid load	bicarb buffer	lungs → ↓ PCO ₂	kidney → ↑ HCO ₃ ⁻

Three speeds, always. Different effectors, same architecture.

Connect Santiago to Oscar

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ACTIVITY

Santiago's kidneys failed last week. Predict his acid-base status, without being told.

Failed kidneys → can't excrete H^+ or generate new bicarbonate →

UREMIC METABOLIC ACIDOSIS

(the U in MUDPILES — opposite end of the matrix from Oscar)

Same machinery. Two patients. Opposite directions. That is the framework.

(MUDPILES is a common medical mnemonic used to recall the differential diagnosis for metabolic acidosis. It helps clinicians identify the underlying cause of acidosis, such as toxin ingestion, renal failure, or metabolic dysfunction, and typically includes Methanol, **Uremia**, Diabetic Ketoacidosis, Propylene Glycol/Paracetamol, Iron/Isoniazid, Lactic Acidosis, Ethylene Glycol, and Salicylates.)

Three Takeaways

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CONCEPT

1

Aldosterone exchanges Na^+ for $\text{K}^+ + \text{H}^+$. One hormone explains all four of Oscar's findings.

2

ABG algorithm: pH first, then match against PCO_2 (respiratory) or HCO_3^- (metabolic). 30 seconds.

3

Three speeds, always: instant buffers, fast lungs, slow kidneys. Same architecture as water and sodium.

Midterm — Wednesday, May 13

Practice + review session: Monday, May 11. Room 8403.

Format: MCQ + short answer + 1–2 clinical-reasoning prompts. Covers Weeks 0–4.

Three things to do this weekend:

1. Re-do every ? from Weeks 0–4 without notes.
2. Work the Pearson IP2 Urine Concentration & Volume review.
3. Draw the 2×2 acid-base matrix from memory until it's automatic.